Cole Liu

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A curious and efficient student with a passion for optimization. Motivated by a strong sense of ownership of work. Driven to apply my logical problem-solving skills to develop innovative solutions for complex challenges.

Education

Arizona State University, GPA 3.83 — Aug. 2022 – May 2026
(In Progress) BS Computer Science, BS Mathematics — *Tempe, AZ*

Experience

Software Engineering Intern — Aug. 2023 – Aug. 2025
Popl — *Remote*

- Work on **full-stack implementation** to support Popl Teams using **React** and **Node.js**
- Authored and maintained comprehensive unit tests to ensure code quality and reliability
- Supported the integration of CRM platforms for lead management

Undergraduate Research Assistant — Jan. 2024 – May 2024
Arizona State University — *Tempe, AZ*

- Developed a machine learning regression model for lithium-ion battery capacity degradation using **Python**
- Processed and validated real-world EIS sensor data from physical battery systems to ensure model accuracy.
- Enabled instantaneous prediction of battery state of health without requiring EIS evaluation
- Required data mining of **large datasets** in **Pandas**

Undergraduate Research Fellow — May 2024 – Aug. 2024
University of Illinois Urbana-Champaign — *Champaign, IL*

- National Science Foundation REU Recipient
- NSF Engineering Research Center - Power Optimization of Electro-Thermal Systems
  - Interdisciplinary research group at UIUC, Stanford, Howard, and University of Arkansas
- Generated a framework for modeling the life-cycle cost of engineering systems, integrated with **Spi2Py**
- Created a **Graph Neural Network** regression architecture to optimize complex engineering systems
- Presented findings at 2024 Illinois Summer Research Symposium (ISRS) and STEM Career Symposium

Projects

Graph Neural Network for Design Optimization | *Python, PyTorch, CUDA* — May 2024 – Aug. 2024

- Developed a Machine Learning model that leverages graph features for more accurate predictions
- Ranked a **dataset** of 43,249 analog circuits for desired metrics
- Reduced required computation by 99.8%, evaluated in 4.8 seconds vs 10 hours

3D Pathfinding for Robotic Arms (Pre Publication) | ***C++**, Python, Algorithms* — May 2024 – Apr. 2025

- Developed a pathfinding algorithm in **C++** to navigate robotic arms through densely packed environments
- Modeled spatial systems using 3D matrices and leveraged **Fast Fourier Transforms** to reduce runtime from $O(n^2)$ to $O(n \log n)$, enabling rapid evaluation of **large-scale systems**
- Generated an enumeration framework of collision-free trajectories, providing the foundational dataset required to train downstream deep learning models on system life-cycle costs

Technical Skills

Languages: **JS, HTML/CSS, SQL, Python, C/C++**, Java, C#, R, CLISP, Swift, Scheme, Prolog, MATLAB
Frameworks: **React, JUnit, Agile**, Tensorflow, PyTorch, Scikit-learn, CUDA, SwiftUI
Developer Tools: **Git, Jira, Docker**, Linux, Bash, Unix, VS Code, PyCharm, Claude, ChatGPT, Gemini
AI & LLM Ecosystem: **OpenAI/Anthropic APIs, MCP, LangChain, RAG**, Hugging Face, Vector DBs

Relevant Coursework

Intro to Software Engineering, **Data Structures and Algorithms**, **Probability and Statistics**, **Operating Systems**, Intro to Cybersecurity, Intro Theoretical Comp Sci, Applied Linear Algebra, Advanced Calculus, Statistical Model Inference, Assembly Language Programming, Numerical Analysis, Applied Statistics, Discrete Math Structures, Modern Differential Equations, Foundations of Machine Learning